10TH CLASS MATHEMATICS — NEW SYLLABUS (PECTAA)

Exercise 6.2 — Addition and Subtraction of Vectors

Unit 6 | Solved Exercise | New Book (PECTAA)

Exercise 6.2 (Solved)

Add or subtract vectors component by component: the i parts go together and the j parts go together. For a magnitude, use $|a| = \sqrt{(x^2 + y^2)}$.

1. If $a = 7i - 3j$ and $b = i + 5j$, then find the following vectors.

(i) a + b (ii) a + 3b (iii) 3a + ½b

(iv) b − a (v) 4b − 5a (vi) 1.5a − b

(i) a + b

$a + b = (7\bar{i} - 3\bar{j}) + (\bar{i} + 5\bar{j})$

$= (7 + 1)\bar{i} + (-3 + 5)\bar{j}$

Answer: $a + b = 8\bar{i} + 2\bar{j}$

(ii) a + 3b

$a + 3b = (7\bar{i} - 3\bar{j}) + 3(\bar{i} + 5\bar{j})$

$= (7 + 3)\bar{i} + (-3 + 15)\bar{j}$

Answer: $a + 3b = 10\bar{i} + 12\bar{j}$

(iii) 3a + ½b

$3a + \frac{1}{2} b = 3(7\bar{i} - 3\bar{j}) + \frac{1}{2}(\bar{i} + 5\bar{j})$

$= (21\bar{i} - 9\bar{j}) + (\frac{1}{2}\bar{i} + \frac{5}{2}\bar{j})$

$= (21 + \frac{1}{2})\bar{i} + (-9 + \frac{5}{2})\bar{j}$

Answer: $3a + \frac{1}{2} b = \frac{43}{2}\bar{i} - \frac{13}{2}\bar{j}$

(iv) b − a

$b - a = (\bar{i} + 5\bar{j}) - (7\bar{i} - 3\bar{j})$

$= (1 - 7)\bar{i} + (5 + 3)\bar{j}$

Answer: $b - a = -6\bar{i} + 8\bar{j}$

(v) 4b − 5a

$4b - 5a = 4(\bar{i} + 5\bar{j}) - 5(7\bar{i} - 3\bar{j})$

$= (4 - 35)\bar{i} + (20 + 15)\bar{j}$

Answer: $4b - 5a = -31\bar{i} + 35\bar{j}$

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(vi) 1.5a − b

$\frac{3}{2}$ a − b = $\frac{3}{2}$ $(7\bar{i} - 3\bar{j}) - (\bar{i} + 5\bar{j})$

$= (\frac{21}{2} - 1)\bar{i} + (-\frac{9}{2} - 5)\bar{j}$

Answer: $\frac{3}{2}$ **a − b =** $\frac{19}{2}$ $\bar{i}$ − $\frac{19}{2}$ $\bar{j}$

2. If a = 6i − j, b = i + 5j and c = 3i + 5j, then find the magnitudes of the following vectors.

(i) b − c (ii) a − 2b + c (iii) c − b − a

(i) b − c

$b - c = (\bar{i} + 5\bar{j}) - (3\bar{i} + 5\bar{j}) = -2\bar{i}$

$|b - c| = \sqrt{((-2)^2 + 0^2)} = 2$

Answer: |b − c| = 2

(ii) a − 2b + c

$a - 2b + c = (6\bar{i} - \bar{j}) - 2(\bar{i} + 5\bar{j}) + (3\bar{i} + 5\bar{j})$

$= (6 - 2 + 3)\bar{i} + (-1 - 10 + 5)\bar{j}$

$= 7\bar{i} - 6\bar{j}$

$|a - 2b + c| = \sqrt{((7)^2 + (-6)^2)} = \sqrt{(49 + 36)}$

Answer: |a − 2b + c| = √85

(iii) c − b − a

$c - b - a = (3\bar{i} + 5\bar{j}) - (\bar{i} + 5\bar{j}) - (6\bar{i} - \bar{j})$

$= (3 - 1 - 6)\bar{i} + (5 - 5 + 1)\bar{j}$

$= -4\bar{i} + \bar{j}$

$|c - b - a| = \sqrt{((-4)^2 + 1^2)} = \sqrt{(16 + 1)}$

Answer: |c − b − a| = √17

3. Find the values of x and y in each of the following equations.

(i) (xi + yj) + (2i + 3j) = 7i + 6j

$(x + 2)\bar{i} + (y + 3)\bar{j} = 7\bar{i} + 6\bar{j}$

Comparing the $\bar{i}$ parts: x + 2 = 7

Comparing the $\bar{j}$ parts: y + 3 = 6

x = 5 and y = 3

Answer: x = 5, y = 3

(ii) (xi − 5j) + (3i + 5j) = −8i + yj

$(x + 3)\bar{i} + 0\bar{j} = -8\bar{i} + y\bar{j}$

x + 3 = −8

x = −11

y = 0

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Answer: x = −11, y = 0

(iii) (yi + 3j) + (−5i + 2xj) = 9i + 7j

$(y - 5)\bar{i} + (3 + 2x)\bar{j} = 9\bar{i} + 7\bar{j}$

$y - 5 = 9$

$y = 14$

$3 + 2x = 7$

$2x = 4$

$x = 2$

Answer: x = 2, y = 14

4. If a = i + 3j, c = 2i + j and a + 2b = c, then find |b|.

$a + 2b = c \implies 2b = c - a$

$2b = (2\bar{i} + \bar{j}) - (\bar{i} + 3\bar{j})$

$= (2 - 1)\bar{i} + (1 - 3)\bar{j}$

$= \bar{i} - 2\bar{j}$

$b = \frac{1}{2}\bar{i} - \bar{j}$

$|b| = \sqrt{(\frac{1}{2})^2 + (-1)^2} = \sqrt{\frac{1}{4} + 1} = \sqrt{\frac{5}{4}}$

Answer: |b| = √5/2

5. If a = −2i + 7j and b = 3i − 5j, then find the vector components of a + 5b.

$a + 5b = (-2\bar{i} + 7\bar{j}) + 5(3\bar{i} - 5\bar{j})$

$= (-2 + 15)\bar{i} + (7 - 25)\bar{j}$

$= 13\bar{i} - 18\bar{j}$

So the vector components of a + 5b are $13\bar{i}$ and $-18\bar{j}$.

Answer: 13i and −18j

6. If 5i − 3j = m(i − 10j) + n(4i − 3j), then find the values of m and n.

$m(\bar{i} - 10\bar{j}) + n(4\bar{i} - 3\bar{j}) = 5\bar{i} - 3\bar{j}$

$(m + 4n)\bar{i} + (-10m - 3n)\bar{j} = 5\bar{i} - 3\bar{j}$

Comparing the $\bar{i}$ parts: $m + 4n = 5$ ... (i)

Comparing the $\bar{j}$ parts: $-10m - 3n = -3$ ... (ii)

From (i): $m = 5 - 4n$

Put this in (ii): $-10(5 - 4n) - 3n = -3$

$-50 + 40n - 3n = -3$

$37n = 47$

$n = 47/37$

$m = 5 - 4(\frac{47}{37}) = \frac{185 - 188}{37} = -\frac{3}{37}$

Answer: m = −3/37, n = 47/37

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